

PAGM — Prediction Aggregation Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Collective Prediction Governance Standard (CPGS)

Operational Layer: Collective Prediction Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Aggregation

Version: 1.0

Status: Canonical · Open Module

Effective Date: 9 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Aggregation

Canonical Definition

Prediction Aggregation Module (PAGM) defines the governance conditions under which multiple predictive contributions may be selected, normalized, segmented, weighted, combined, or otherwise composed into an aggregated predictive representation without obscuring material differences in Prediction Object, methodology, Prediction Basis, assumptions, Probability & Uncertainty, feasibility, horizon, boundary, source diversity, dependence, or predictive context.

PAGM governs the transformation:



Governed Predictive Contributions + Governed Predictive Source Diversity Profile + Governed Prediction Dependence & Variation Profile + Defined Aggregation Method

into:

Governed Prediction Aggregation

It establishes whether predictive contributions are sufficiently compatible for aggregation, what aggregation method is being applied, how contribution weights are assigned, what information may be lost through aggregation, and what qualifications must accompany the aggregated result.

Its foundational principle is:

Predictions that can be mathematically combined are not necessarily predictions that can be methodologically aggregated.

Operational Role

PAGM is the third internal governance space of the Collective Prediction Governance Standard (CPGS) .

PSDM establishes:

what predictive source diversity exists.

PDVM establishes:

how independent, dependent, convergent, divergent, or structurally related the predictive contributions are.

PAGM establishes:

whether and how those predictive contributions may legitimately be aggregated.

The focused operational question is:

Which predictive contributions may legitimately be combined, under what aggregation method, with what weighting, and subject to what methodological qualifications?

The primary output is:

Governed Prediction Aggregation

Internal Module Flow

Predictive Source Diversity → Prediction Dependence & Variation → Prediction Aggregation → Predictive Influence & Concentration → Collective Prediction Representation

Current Position: Prediction Aggregation

Prediction Aggregation

Prediction Aggregation is the governed process through which multiple predictive contributions are combined into an aggregated predictive representation.

Aggregation may include:

arithmetic averaging; median composition; weighted averaging; probability pooling; expert aggregation; ensemble composition; voting; market-derived aggregation; Bayesian combination; hierarchical composition; rule-based composition; other defined aggregation methods.

Aggregation ≠ Collective Legitimacy

PAGM governs composition.

Final collective prediction legitimacy is established only after the broader CP GS governance structure is completed.

Aggregation Eligibility

Before contributions are aggregated, their eligibility for aggregation must be established.

Available Prediction ≠ Aggregation-Eligible Prediction

A contribution may exist while being unsuitable for direct aggregation.

Aggregation Compatibility

Aggregation Compatibility concerns whether predictive contributions are sufficiently aligned in meaning and governance context to be legitimately combined.

Numerical Compatibility ≠ Methodological Compatibility

Two predictions may use the same numerical format while representing materially different predictive objects or conditions.

Prediction Object Compatibility

Contributing predictions should refer to sufficiently compatible Prediction Objects.

Similar Question ≠ Same Prediction Object

Outcome Definition Compatibility

Predictions should use compatible outcome definitions.

Example:

Company launches product

may not mean the same thing as:

Company begins commercial sales in all target markets.

Similar Outcome Language ≠ Same Governed Outcome

Horizon Compatibility

Predictive contributions should use compatible Prediction Horizons unless the aggregation method explicitly governs horizon differences.

Probability at Horizon A ≠ Probability at Horizon B

Boundary Compatibility

Predictive boundaries may differ across:

geography; jurisdiction; population; organization; operational scope; market; system.

Different Boundary ≠ Directly Aggregable Prediction by Default

Methodological Compatibility

Predictions formed through different methodologies may sometimes be legitimately aggregated.

The differences must remain understood.

Different Methodologies ≠ Incompatible by Default

But:

Methodological Difference $\neq$ Irrelevant Difference

Prediction Basis Compatibility

Contributions may rely upon different or overlapping Prediction Bases.

PAGM does not govern those bases themselves.

It preserves their relevance to aggregation.

Shared Prediction Basis $\neq$ Independent Aggregation Weight

Assumption Compatibility

Predictions may contain materially different assumptions.

Conflicting Assumptions $\neq$ Directly Aggregable Predictions by Default

Probability Representation Compatibility

Predictions may express probability differently.

Examples include:

point probability; probability interval; qualitative likelihood; distribution; confidence category.

Same Numerical Scale $\neq$ Same Probability Meaning

PPUS owns probability and uncertainty governance.

PAGM governs whether representations may be compositionally combined.

Feasibility Compatibility

Predictions may carry different feasibility states.

Feasible Prediction + Infeasible Prediction $\neq$ Neutral Average by Default

Lifecycle Compatibility

A current prediction should not automatically be aggregated with:

withdrawn predictions; superseded predictions; obsolete predictions; predictions formed under materially different conditions.

Superseded Prediction $\neq$ Current Aggregation Input by Default

Temporal Compatibility

Predictions formed at materially different times may reflect different information states.

Older Prediction $\neq$ Current Prediction

Aggregation Set

An Aggregation Set is the defined collection of predictive contributions eligible for a specific aggregation operation.

Available Contributions $\neq$ Aggregation Set

Aggregation Set Boundary

The aggregation set should define which predictions are included and excluded.

Inclusion Rule

An Inclusion Rule establishes conditions under which a prediction enters the aggregation set.

Exclusion Rule

An Exclusion Rule establishes conditions under which a prediction does not enter or is removed from the aggregation set.

Exclusion $\neq$ Invalid Prediction

A legitimate prediction may still be incompatible with a particular aggregation.

Aggregation Segmentation

Predictions that cannot legitimately be combined may be separated into distinct aggregation groups.

Segmentation may occur by:

Prediction Object; horizon; methodology; assumptions; boundary; source type; information condition; feasibility state.

Segmentation $\neq$ Aggregation Failure

Segmentation may be the methodologically correct result.

Forced Aggregation

Forced Aggregation occurs where materially incompatible predictions are combined merely because a single collective number is desired.

Single Number Desired $\neq$ Single Number Legitimate

Aggregation Method

The Aggregation Method defines how eligible predictive contributions are transformed into an aggregated result.

Aggregation Method Must Be Explicit

A governed aggregation should not conceal the method used.

Opaque Aggregation $\neq$ Governed Aggregation

Arithmetic Mean

An arithmetic mean assigns equal numerical weight to each included contribution.

Equal Numerical Weight $\neq$ Equal Predictive Relevance

Median Aggregation

Median aggregation may reduce sensitivity to extreme values.

Median $\neq$ Methodologically Neutral Result

Weighted Aggregation

Weighted Aggregation assigns different influence to predictive contributions.

Weight $\neq$ Truth

Weighting Basis

Weights may be based upon governed factors such as:

methodology; historical performance; calibration; source relevance; information distinctiveness; dependence; timeliness; domain expertise; feasibility context.

Weighting Factor $\neq$ Automatic Weighting Rule

The methodology must define how factors affect weights.

Equal Weighting

Equal weighting may be legitimate where methodology supports equal treatment.

Equal Weighting $\neq$ Absence of Methodology

Choosing equal weighting is itself a methodological decision.

Performance Weighting

Historical predictive performance may influence contribution weight.

Past Accuracy $\neq$ Permanent Predictive Weight

Performance Weighting Requires Context

Historical performance may be affected by:

prediction domain; horizon; information access; Predictor–Reality Proximity; number of predictions; market conditions.

Observed Accuracy $\neq$ Demonstrated Predictive Skill

Expertise Weighting

Expertise may influence aggregation.

Expert Status $\neq$ Predictive Accuracy by Default

Source Diversity Weighting

Source diversity may inform aggregation structure.

Source Diversity $\neq$ Automatic Higher Weight

Dependence-Adjusted Weighting

Where contributions are materially dependent, aggregation may account for their shared structure.

Ten Dependent Predictions $\neq$ Ten Independent Units of Predictive Information

Dependence Adjustment

Dependence adjustment may reduce the effective contribution of duplicated or strongly related predictive information.

Dependence Adjustment ≠ Prediction Exclusion by Default

Effective Contribution Weight

Effective Contribution Weight represents the influence a contribution carries after applicable aggregation rules and dependence adjustments.

Nominal Weight ≠ Effective Weight

Weight Normalization

Weights may require normalization to produce a coherent aggregation structure.

Normalized Weight ≠ Methodological Legitimacy

Weight Cap

A methodology may limit the maximum weight assigned to one contribution or source class.

Weight Cap ≠ Elimination of Influence Concentration

Influence concentration is governed further in the next CPGS layer.

Dynamic Weighting

Weights may change over time.

Current Weight ≠ Historical Weight

Material changes should remain traceable.

Weight Versioning

Historical aggregation should preserve the weights applicable when the aggregate was produced.

Later Weighting Method ≠ Historical Weighting Method

Probability Pooling

Probability Pooling combines multiple governed probability representations.

Probability Pooling ≠ Probability Validation

PPUS governs the legitimacy of individual probability representations.

Linear Pooling

A weighted arithmetic combination may be used where methodologically appropriate.

Linear Pooling ≠ Universal Aggregation Method

Geometric Pooling

A geometric or logarithmic combination may be used where methodology supports it.

Mathematical Sophistication ≠ Governance Legitimacy

Bayesian Combination

Bayesian methods may combine predictive information under defined assumptions.

Bayesian Method ≠ Automatic Independence

Independence assumptions must remain governed.

Ensemble Aggregation

AI or statistical models may be combined through ensemble methods.

Model Ensemble ≠ Independent Predictive Collective by Default

Voting Aggregation

Predictions may be combined through voting.

Majority Vote ≠ Probability Estimate

Majority ≠ Predictive Truth

Expert Aggregation

Expert judgments may be aggregated.

Expert Consensus ≠ Independent Confirmation

Market-Derived Aggregation

Prediction markets may produce market-derived collective signals.

Market Price ≠ Governed Probability Estimate

A market price may require methodological transformation before being represented as probability.

Market Aggregation ≠ Population Aggregation

Liquidity Condition

Where market-derived aggregation is used, liquidity may materially affect interpretation.

High Liquidity ≠ Methodological Legitimacy

Capital-Weighted Aggregation

Markets may effectively weight participants through capital deployment.

Capital Weight ≠ Predictive Merit

Strategic Participation

Market participants may act for reasons other than predictive belief.

Trade ≠ Pure Prediction Contribution

Hierarchical Aggregation

Predictions may first be aggregated within groups and then across groups.

Example:

Regional Forecasts → Regional Aggregates → Global Aggregate

Aggregate of Aggregates ≠ Raw Prediction Aggregate

The hierarchical structure must remain traceable.

Aggregation Layer

Each aggregation step should remain identifiable.

Nested Aggregation

A predictive contribution may already represent an aggregate.

Aggregate Input ≠ Individual Prediction

Double Counting

Double Counting occurs where the same predictive information enters aggregation multiple times without appropriate recognition.

Repeated Information ≠ Additional Predictive Evidence

Upstream Aggregate Duplication

A system may aggregate both an upstream collective forecast and the individual forecasts already contained within it.

Aggregate + Its Components ≠ Independent Contributions

Prediction Deduplication

Where duplicated predictive contributions are identified, aggregation should prevent unintended repeated influence.

Deduplication ≠ Prediction Deletion

The prediction remains historically identifiable even if excluded from a particular aggregation.

Aggregation Order

For some methodologies, aggregation order may affect the result.

Different Aggregation Order ≠ Same Result by Default

Aggregation Sensitivity

Aggregation Sensitivity represents how materially the collective result changes when reasonable aggregation choices change.

Relevant choices may include:

inclusion; exclusion; weighting; segmentation; method; dependence adjustment.

Low Aggregation Sensitivity

Reasonable methodological variation produces similar aggregate results.

High Aggregation Sensitivity

Reasonable methodological variation produces materially different aggregate results.

Stable Aggregate ≠ Accurate Prediction

Sensitive Aggregate ≠ Invalid Aggregate

Sensitivity is a governance characteristic.

Aggregation Robustness

Aggregation Robustness exists where the collective representation remains materially stable across reasonable governance-compatible aggregation methods.

Robust Aggregation ≠ Correct Prediction

Aggregation Fragility

Aggregation Fragility exists where small changes in inclusion, weighting, or method materially alter the result.

Aggregation Fragility ≠ Prediction Uncertainty

Contributor Removal Sensitivity

An aggregate may depend heavily upon one or a small number of contributions.

One-Contributor Removal Changes Result Materially → Aggregation Concentration Signal

PAGM preserves this condition for subsequent influence governance.

Method Selection

The selected aggregation method should be appropriate to:

Prediction Object; contribution type; probability representation; source structure; dependence structure; intended collective representation.

Convenient Method ≠ Appropriate Method

Method Selection Rationale

The reason for selecting an aggregation method should remain traceable.

Method Selection ≠ Outcome Optimization

A method should not be chosen merely because it produces a preferred collective result.

Result-Driven Method Selection

Selecting aggregation rules after observing which method produces the desired output creates governance risk.

Preferred Result ≠ Legitimate Aggregation Method

Aggregation Method Lock

Where appropriate, the aggregation method should be established before the relevant outcome becomes known.

Ex Ante Method ≠ Ex Post Method Selection

Aggregation Transparency

A governed aggregate should preserve sufficient information to reconstruct:

included contributions; excluded contributions; aggregation method; weights; dependence adjustments; segmentation; resulting aggregate.

Aggregate Number Alone ≠ Governed Aggregation

Aggregation Rationale

The aggregation should remain explainable at the level required by its use.

Explainability ≠ Simplification

A complex aggregation method may remain legitimate if sufficiently documented.

Aggregation Qualification

Material limitations should accompany the aggregated result.

Examples include:

High Source Dependence

Limited Representation

Material Horizon Variation

High Aggregation Sensitivity

Market Concentration

Qualified Aggregate ≠ Weak Aggregate by Default

Qualification preserves meaning.

Aggregation Uncertainty

Where the aggregation method itself introduces material uncertainty, that uncertainty should remain distinguishable from predictive uncertainty.

Aggregation Uncertainty ≠ Prediction Uncertainty

Aggregation Conflict

Different legitimate aggregation methods may produce materially different results.

Aggregation Conflict ≠ Average the Aggregates by Default

Methodological Plurality

Where no single aggregation method is sufficiently dominant, multiple governed aggregates may be preserved.

Multiple Governed Aggregates ≠ Governance Failure

Aggregation State

PAGM may classify an aggregation as:

Aggregation Eligible

Conditionally Aggregation Eligible

Segmentation Required

Aggregation Restricted

Methodologically Incompatible

Aggregation Uncertain

Unable to Determine

Aggregation Eligible

Contributions are sufficiently compatible for the defined aggregation.

Conditionally Aggregation Eligible

Aggregation is legitimate subject to explicit qualifications or adjustments.

Segmentation Required

Contributions require separation before aggregation.

Aggregation Restricted

Only a subset or restricted form of aggregation is legitimate.

Methodologically Incompatible

Material differences prevent legitimate direct aggregation under the proposed method.

Aggregation Uncertain

Material methodological questions remain unresolved.

Unable to Determine

Available governance information is insufficient to establish aggregation legitimacy.

Unable to Determine $\neq$ Aggregation Eligible

Aggregated Predictive Representation

The output of aggregation is an intermediate predictive representation.

Aggregated Predictive Representation $\neq$ Governed Collective Prediction

Further CPGS governance remains required.

Aggregation Version

Every material aggregation should preserve:

input set; method; weights; adjustments; qualifications; output; timestamp/version.

Current Aggregate $\neq$ Historical Aggregate

Reaggregation

Changes in predictive contributions or aggregation methodology may produce a new aggregate.

Reaggregation $\neq$ Historical Replacement

Later Aggregation Method $\neq$ Historical Aggregation Method

Relationship to PSDM

PSDM establishes source diversity.

PAGM consumes that structure when determining aggregation eligibility and composition.

Source Diversity ≠ Aggregation Weight

Relationship to PDVM

PDVM establishes dependence and variation.

PAGM uses those outputs to prevent dependent contributions from being treated automatically as equivalent independent information.

Dependence ≠ Automatic Exclusion

Relationship to PICM

PAGM establishes aggregation structure and contribution weights.

PICM subsequently governs:

Predictive Influence & Concentration

including whether the aggregate is materially dominated by particular contributors, clusters, capital positions, models, or information origins.

Weight ≠ Legitimate Influence by Default

Relationship to CPRM

PAGM does not establish final collective representation.

CPRM governs whether the resulting collective structure may legitimately be represented as a Governed Collective Prediction.

Aggregation ≠ Representation

Relationship to PPUS

PPUS governs Probability & Uncertainty.

PAGM may aggregate governed probability representations but does not redefine their meaning.

Aggregation ≠ Probability Governance Replacement

Relationship to PBAS

PBAS governs Prediction Basis.

PAGM may account for basis compatibility and overlap without governing the underlying basis itself.

Relationship to PAS

PAS governs Predictive Assumptions.

PAGM may account for assumption compatibility without redefining those assumptions.

Relationship to PFES

PFES governs Prediction Feasibility.

PAGM may preserve feasibility differences when determining aggregation compatibility.

Relationship to POFPS

POFPS later governs Prediction Outcome & Performance.

Historical aggregation methodology must remain available so collective performance can be interpreted against the actual predictive structure that existed at formation.

Governance Requirements

PAGM requires that:

1. Prediction Aggregation remain distinguishable from collective legitimacy; 2. aggregation eligibility be established before direct composition; 3. available predictions not automatically become aggregation inputs; 4. Prediction Object compatibility remain assessable; 5. similar questions not automatically be treated as the same Prediction Object; 6. outcome definitions remain compatible or explicitly qualified; 7. horizon differences remain governed; 8. boundary differences remain governed; 9. methodological differences remain identifiable; 10. different methodologies not automatically be treated as incompatible; 11. Prediction Basis relationships remain preservable where material; 12. Shared Prediction Basis not automatically create independent aggregation weight; 13. conflicting assumptions remain identifiable; 14. probability representations remain compositionally interpretable; 15. same numerical scale not establish equivalent probability meaning; 16. feasibility differences remain preservable; 17. lifecycle states remain relevant to aggregation eligibility; 18. superseded predictions not enter current aggregation by default; 19. temporal differences remain identifiable; 20. the Aggregation Set remain explicit; 21. inclusion rules remain identifiable; 22. exclusion rules remain identifiable; 23. exclusion from aggregation not imply invalidity of the prediction; 24. segmentation remain available where direct aggregation is illegitimate; 25. forced aggregation be avoided; 26. demand for one number not override methodological incompatibility; 27. aggregation method remain explicit; 28. opaque aggregation not qualify as governed aggregation; 29. equal numerical weight remain distinguishable from equal predictive relevance; 30. weighting basis remain explicit; 31. weighting factors remain distinguishable from weighting rules; 32. equal weighting remain recognized as a methodological choice; 33. historical performance weighting remain contextualized; 34. past accuracy not create permanent predictive weight; 35. observed accuracy remain distinguishable from demonstrated predictive skill; 36. expertise weighting remain governable; 37. expert status not automatically establish predictive accuracy; 38. source diversity not automatically create higher weight; 39. dependence-adjusted weighting remain possible; 40. dependent contributions not automatically be counted as independent information units; 41. dependence adjustment remain distinguishable from exclusion; 42. nominal weight remain distinguishable from effective weight; 43. weight normalization not establish methodological legitimacy;
44. weight caps remain distinguishable from full influence governance; 45. dynamic weighting changes remain traceable; 46. historical weights remain reconstructable; 47. probability pooling remain distinct from probability validation; 48. mathematical sophistication not substitute for governance legitimacy; 49. Bayesian aggregation not assume independence by default; 50. ensemble aggregation not assume model independence; 51. majority voting remain distinguishable from probability estimation; 52. expert consensus remain distinguishable from independent confirmation; 53. market price remain distinguishable from governed probability; 54. market aggregation remain distinguishable from population aggregation; 55. capital weight remain distinguishable from predictive merit; 56. strategic participation remain representable where material; 57. hierarchical aggregation remain traceable; 58. nested aggregates remain distinguishable from individual predictions; 59. Double Counting remain identifiable; 60. upstream aggregate duplication remain preventable; 61. deduplication remain distinguishable from prediction deletion; 62. aggregation order remain governable where material; 63. Aggregation Sensitivity remain assessable; 64. aggregation robustness remain distinguishable from predictive accuracy; 65. aggregation fragility remain distinguishable from Prediction Uncertainty; 66. contributor-removal sensitivity remain identifiable; 67. method selection remain appropriate to the governed predictive structure; 68.

convenience not substitute for methodological appropriateness; 69. method-selection rationale remain traceable; 70. aggregation method not be selected merely to produce a preferred result; 71. ex ante and ex post method selection remain distinguishable; 72. aggregation inputs and transformations remain reconstructable; 73. aggregate number alone not qualify as governed aggregation; 74. material aggregation qualifications remain attached to the result; 75. Aggregation Uncertainty remain distinguishable from Prediction Uncertainty; 76. aggregation conflicts remain preservable; 77. multiple legitimate aggregation methods not be forcibly collapsed; 78. aggregation states remain explicit; 79. Unable to Determine remain available as a legitimate state; 80. Aggregated Predictive Representation remain distinguishable from Governed Collective Prediction; 81. aggregation versions remain historically reconstructable; 82. reaggregation not erase prior aggregation states; 83. source diversity remain distinguishable from aggregation weight; 84. dependence remain distinguishable from automatic exclusion; 85. aggregation remain distinguishable from Predictive Influence & Concentration; 86. aggregation remain distinguishable from final Collective Prediction Representation.

Governance Logic

PAGM applies the following governance logic:

Governed Predictive Source Diversity Profile

Governed Prediction Dependence & Variation Profile

Contributing Governed Predictions

↓

Candidate Aggregation Set Identified

↓

Prediction Object Compatibility Assessed

↓

Outcome Definition Compatibility Assessed

↓

Prediction Horizon Compatibility Assessed

↓

Predictive Boundary Compatibility Assessed

↓

Methodological Compatibility Assessed

↓

Prediction Basis Relationships Considered

↓

Predictive Assumption Compatibility Assessed

↓

Probability Representation Compatibility Assessed

↓

Feasibility Compatibility Assessed

↓

Lifecycle / Temporal Compatibility Assessed

↓

Aggregation Eligibility Established

↓

Segmentation Applied where required

↓

Aggregation Set Established

↓

Aggregation Method Selected

↓

Method Selection Rationale Preserved

↓

Weighting Basis Established where applicable

↓

Dependence Adjustment Applied where required

↓

Duplicate / Nested Contributions Controlled

↓

Effective Contribution Weights Established

↓

Aggregation Performed

↓

Aggregation Sensitivity / Robustness Assessed where material

↓

Aggregation Qualifications Established

↓

Aggregation State Established

↓

Historical Aggregation Version Preserved

↓

Governed Prediction Aggregation

↓

Predictive Influence & Concentration

Where legitimate aggregation cannot be established:

Prediction Aggregation Governance Gap → Compatibility Clarification / Segmentation / Method Revision /
Weighting Revision / Aggregation Restriction / Methodologically Incompatible / Aggregation Uncertain /
Unable to Determine

Minimum Implementation Framework

A minimum implementation of PAGM must establish:

A. Collective Prediction Reference

The candidate collective prediction to which the aggregation belongs.

B. Aggregation Set

The predictive contributions eligible for the specific aggregation.

C. Inclusion / Exclusion Rule

The conditions governing participation in the aggregation.

D. Compatibility State

The relevant Prediction Object, horizon, boundary, methodology, assumption, probability, and feasibility compatibility.

E. Segmentation State

Any required separation of predictive contributions.

F. Aggregation Method

The defined method used to combine contributions.

G. Method Selection Rationale

Why the aggregation method is appropriate.

H. Weighting Basis

Where weighted aggregation is used.

I. Contribution Weight

The nominal weight assigned to each included contribution.

J. Dependence Adjustment

Any adjustment arising from governed predictive dependence.

K. Effective Contribution Weight

The contribution's effective influence within the aggregation.

L. Aggregated Predictive Representation

The resulting aggregated output.

M. Aggregation Qualification

Any material limitation, sensitivity, incompatibility, or conditionality.

N. Aggregation State

At minimum:

Aggregation Eligible / Conditionally Aggregation Eligible / Segmentation Required / Aggregation Restricted / Methodologically Incompatible / Aggregation Uncertain / Unable to Determine

O. Historical Aggregation Version

The reconstructable inputs, method, weights, adjustments, result, and version applicable when the aggregate was produced.

Governance Outputs

PAGM may produce:

Governed Prediction Aggregation; Aggregation Set; Aggregation Eligibility State; Prediction Object Compatibility State; Outcome Definition Compatibility State; Horizon Compatibility State; Boundary Compatibility State; Methodological Compatibility State; Prediction Basis Compatibility Record; Assumption Compatibility State;

Probability Representation Compatibility State; Feasibility Compatibility State; Lifecycle Compatibility State; Inclusion Rule; Exclusion Rule; Segmentation Requirement; Aggregation Segment; Aggregation Method; Aggregation Method Rationale; Weighting Basis; Contribution Weight; Dependence Adjustment; Effective Contribution Weight; Weight Normalization Record; Weight Cap; Probability Pool; Ensemble Aggregation; Expert Aggregation; Market-Derived Aggregate; Hierarchical Aggregation Structure; Nested Aggregate Record; Double-Counting Record; Deduplication Record; Aggregation Sensitivity State; Aggregation Robustness State; Aggregation Fragility State; Contributor Removal Sensitivity; Aggregation Qualification; Aggregation Uncertainty Record; Aggregation Conflict Record; Aggregation State; Aggregated Predictive Representation; Prediction Aggregation Governance Gap; historical aggregation state; Aggregation Version.

The primary governed output is:

Governed Prediction Aggregation

which becomes an input to:

Predictive Influence & Concentration.

Critical Distinctions

Predictions That Can Be Mathematically Combined Are Not Necessarily Predictions That Can Be Methodologically Aggregated

Aggregation $\neq$ Collective Legitimacy

Available Prediction $\neq$ Aggregation-Eligible Prediction

Numerical Compatibility $\neq$ Methodological Compatibility

Similar Question $\neq$ Same Prediction Object

Similar Outcome Language $\neq$ Same Governed Outcome

Probability at Horizon A $\neq$ Probability at Horizon B

Different Boundary $\neq$ Directly Aggregable Prediction by Default

Different Methodologies $\neq$ Incompatible by Default

Methodological Difference $\neq$ Irrelevant Difference

Shared Prediction Basis $\neq$ Independent Aggregation Weight

Conflicting Assumptions $\neq$ Directly Aggregable Predictions by Default

Same Numerical Scale $\neq$ Same Probability Meaning

Feasible Prediction + Infeasible Prediction $\neq$ Neutral Average by Default

Superseded Prediction $\neq$ Current Aggregation Input by Default

Older Prediction $\neq$ Current Prediction

Available Contributions $\neq$ Aggregation Set

Exclusion $\neq$ Invalid Prediction

Segmentation $\neq$ Aggregation Failure

Single Number Desired $\neq$ Single Number Legitimate

Aggregation Method $\neq$ Neutral Mechanism

Opaque Aggregation $\neq$ Governed Aggregation

Equal Numerical Weight $\neq$ Equal Predictive Relevance

Weight $\neq$ Truth

Weighting Factor $\neq$ Automatic Weighting Rule

Equal Weighting $\neq$ Absence of Methodology

Past Accuracy $\neq$ Permanent Predictive Weight

Observed Accuracy $\neq$ Demonstrated Predictive Skill

Expert Status $\neq$ Predictive Accuracy by Default

Source Diversity $\neq$ Automatic Higher Weight

Ten Dependent Predictions $\neq$ Ten Independent Units of Predictive Information

Dependence Adjustment $\neq$ Prediction Exclusion by Default

Nominal Weight $\neq$ Effective Weight

Normalized Weight $\neq$ Methodological Legitimacy

Weight Cap $\neq$ Elimination of Influence Concentration

Current Weight $\neq$ Historical Weight

Probability Pooling $\neq$ Probability Validation

Linear Pooling $\neq$ Universal Aggregation Method

Mathematical Sophistication $\neq$ Governance Legitimacy

Bayesian Method $\neq$ Automatic Independence

Model Ensemble $\neq$ Independent Predictive Collective by Default

Majority Vote $\neq$ Probability Estimate

Majority $\neq$ Predictive Truth

Expert Consensus $\neq$ Independent Confirmation

Market Price $\neq$ Governed Probability Estimate

Market Aggregation $\neq$ Population Aggregation

High Liquidity $\neq$ Methodological Legitimacy

Capital Weight $\neq$ Predictive Merit

Trade $\neq$ Pure Prediction Contribution

Aggregate of Aggregates $\neq$ Raw Prediction Aggregate

Aggregate Input $\neq$ Individual Prediction

Repeated Information $\neq$ Additional Predictive Evidence

Aggregate + Its Components $\neq$ Independent Contributions

Deduplication $\neq$ Prediction Deletion

Different Aggregation Order $\neq$ Same Result by Default

Stable Aggregate $\neq$ Accurate Prediction

Sensitive Aggregate $\neq$ Invalid Aggregate

Robust Aggregation $\neq$ Correct Prediction
Aggregation Fragility $\neq$ Prediction Uncertainty
Convenient Method $\neq$ Appropriate Method
Method Selection $\neq$ Outcome Optimization
Preferred Result $\neq$ Legitimate Aggregation Method
Ex Ante Method $\neq$ Ex Post Method Selection
Aggregate Number Alone $\neq$ Governed Aggregation
Qualified Aggregate $\neq$ Weak Aggregate by Default
Aggregation Uncertainty $\neq$ Prediction Uncertainty
Aggregation Conflict $\neq$ Average the Aggregates by Default
Multiple Governed Aggregates $\neq$ Governance Failure
Unable to Determine $\neq$ Aggregation Eligible
Aggregated Predictive Representation $\neq$ Governed Collective Prediction
Current Aggregate $\neq$ Historical Aggregate
Reaggregation $\neq$ Historical Replacement
Source Diversity $\neq$ Aggregation Weight
Dependence $\neq$ Automatic Exclusion
Weight $\neq$ Legitimate Influence by Default
Aggregation $\neq$ Representation
Aggregation $\neq$ Probability Governance Replacement

Enterprise Application

PAGM enables organizations to combine predictive contributions without allowing the mathematics of aggregation to conceal the governance structure behind the result.

It is applicable to:

prediction markets; expert forecasting; AI model ensembles; multi-agent forecasting; crowdsourced predictions; financial forecasts; institutional forecasting; risk forecasting; geopolitical forecasting; Human-AI prediction systems.

PAGM answers a practical question that appears whenever multiple forecasts are reduced to one result:

Why are these predictions allowed to become one number, and what exactly does that number represent?

Use Case 1 — Three Forecasts Say 70%, but They Predict Different Things

An organization collects three predictions about the launch of a new autonomous vehicle platform.

The forecasts are:

70%

72%

69%

The numbers appear directly aggregable.

However:

Prediction A defines success as:

commercial launch in at least one country within 12 months.

Prediction B defines success as:

commercial launch across the European Union within 18 months.

Prediction C defines success as:

regulatory approval in the United States within 12 months.

The critical governance question is:

Can three similar probability values legitimately be averaged when their Prediction Objects, horizons, boundaries, and outcome definitions are materially different?

PAGM determines that direct aggregation is methodologically incompatible without restructuring or segmentation.

Same Numerical Scale $\neq$ Same Probability Meaning

Single Number Desired $\neq$ Single Number Legitimate

Use Case 2 — 100 Forecasts, but 70 Depend on the Same Upstream Model

A financial institution collects:

100 predictive contributions

about whether a major market index will decline by more than 15% during the next twelve months.

PDVM establishes that:

70 predictions

are materially dependent upon one widely used upstream forecasting model.

The remaining:

30 predictions

represent more independent methodologies and information environments.

A simple arithmetic mean would allow the shared upstream model to influence the collective result approximately seventy times.

The critical governance question is:

Should seventy dependent predictions receive seventy independent units of aggregation weight?

PAGM applies a governed dependence adjustment so the aggregation structure does not mistake repeated information for additional independent predictive evidence.

Ten Dependent Predictions $\neq$ Ten Independent Units of Predictive Information

Repeated Information $\neq$ Additional Predictive Evidence

Canonical Boundary

PAGM governs the eligibility, compatibility, segmentation, selection, weighting, dependence adjustment, composition, sensitivity, qualification, versioning, and historical reconstruction of predictive contributions being aggregated within Collective Prediction Governance. It governs Prediction Object, outcome, horizon, boundary, methodology, assumption, probability-representation, feasibility, lifecycle, and temporal compatibility as they affect aggregation; it governs aggregation sets, inclusion and exclusion rules, aggregation methods, weighting, probability pooling, ensemble composition, market-derived aggregation,

hierarchical aggregation, duplicate control, aggregation sensitivity, robustness, fragility, and methodological conflict. It does not govern Predictive Source Diversity, determine Prediction Dependence itself, validate individual predictions, establish individual Probability & Uncertainty, determine Predictive Influence & Concentration, establish final Collective Prediction legitimacy, govern Prediction Communication & Reliance, determine Operational Reality, or assess predictive performance.

PAGM governs:

whether and how multiple predictive contributions may legitimately be transformed into an aggregated predictive representation.

It does not determine:

whether that aggregated representation may yet be presented as a legitimate Governed Collective Prediction.

Canonical Closing Statement

Aggregation can compress a thousand predictions into one number, but it can also compress away the reasons those predictions were different.

PAGM governs what may be combined, what must remain separate, how contributions are weighted, and which methodological conditions must survive the aggregation process.

The ability to calculate a collective number does not create the authority to call that number a collective prediction.

It

NON-EXECUTABLE NOTICE

OOF® — OriginOpen® Foundation is a non-executable methodological authority.
This website and all publications are reference-only.

CANONICAL LANGUAGE NOTICE

UCL™ is the canonical language layer of OOF® publications.
English is its carrier language.
In case of ambiguity, the published English HTML text prevails.

RIGHTS & PROTECTION

© OOF® — OriginOpen® Foundation
Protected under MIP® — Methodological Intellectual Property.
All rights reserved.



contact@originopenfoundation.org